Detection of proteolytic peptides in protein sequences

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Abstract
Proteins are one of the main biological molecules. They are exposed by digestion with production of short amino acid sequences called peptides which have various implementations in the field of medicine and nutrition improvement. However, even in the largest biological databases peptides are poorly annotated. The state-of-the-art method predicting peptide regions in the protein sequences is DeepPeptide. Hence it outperforms other models solving the same problem, it has several limitations and its evaluation metrics are still only modest in terms of absolute performance. We modified the architecture by switching to another encoder. The ESM-3 and ProtT5 models surpassed the baseline ESM-2 architecture in performance when tested exclusively on human-derived data. This may indicate that more modern approaches perform better on the specialized domain of human data, which is the main area of interest.

1 Introduction

Detection of proteolytic peptides in protein sequences is vital for the number of modern areas of humanity's interests. It develops infant and sports nutrition by identifying bioactive peptides released during digestion that promote health. Dallas et al. [2016], Lönnerdal [2014], Pihlanto [2006], Hernández-Ledesma et al. [2011], Tang et al. [2009], Nongonierma et al. [2015]. It helps develop mRNA vaccines by designing peptide epitopes that elicit strong immune responses and to improve mRNA delivery stability Ahmad et al. [2022], Papadopoulou et al. [2025], Kim et al. [2022], Zhang et al. [2025], Singh et al. [2025]. It is critical for biomarker discovery, as these peptides often reflect disease-specific protease activity and protein degradation in such pathological conditions as cancer, autoimmune and neurodegenerative diseases Klein et al. [2018], Razavi et al. [2021].

Mass spectrometry (MS) is a key tool for studying proteins, but it only detects some of the peptides made when proteins are cut by enzymes. Predicting which peptides will be seen by MS helps researchers focus on the most reliable signals. This saves time, reduces costs, and improves the accuracy of protein identification Serrano et al. [2020], Son et al. [2023].

Several deep learning models now predict peptide detectability Serrano et al. [2020], Son et al. [2023], Yang et al. [2022], Teufel et al. [2022], Guntuboina et al. [2023], Putty Reddy et al. [2023]. However, many of these models work best for a single enzyme, instrument, or data type. A general tool that works across different experiments would be very useful for the proteomics community.

A reliable, easy-to-use peptide detectability predictor has many applications. In targeted proteomics and biomarker studies, it can guide the choice of peptides that serve as markers for diseases Putty Reddy et al. [2023]. In clinical assays, it helps design tests that measure protein levels linked to health or treatment response. In large-scale projects, it can speed up the building of spectral libraries for data-independent acquisition (DIA) and help plan experiments with multiple enzymes Yang et al. [2022]. Overall, better detectability prediction can make proteomics faster, cheaper, and more reliable.

2 Related works

In recent years, researchers have used deep learning to predict which pieces of proteins (peptides) can be seen by mass spectrometry (MS). Serrano et al. [2020] created DeepMSPeptide, an early model that looks only at the amino acid sequence and learns patterns that help it decide if a peptide will be detected. DbyDeep Son et al. [2023] improved on this. Their model uses bidirectional LSTM networks to get better accuracy on different MS machines.

Other methods add information about how proteins are cut before detection. DeepDetect trains two parts together: one part predicts detectability and the other predicts how likely a peptide is to be made by an enzyme digest. This helps find more true peptides in data-independent workflows. DeepPeptide Teufel et al. [2023] focus on where proteins are cut. This model uses conditional random fields to mark cut sites in a protein, giving high accuracy even when reference data are limited.

More recently, transformer-based language models have been applied to peptide tasks. The study Guntuboina et al. [2023] adapted a model called ProtBERT into PeptideBERT. They fine-tuned it to predict several peptide properties, including detectability, solubility, and safety. Finally, PIPER Putty Reddy et al. [2023] used deep-learning–selected peptides, a workflow for measuring protein biomarkers in clinical samples, and showed it works well for detecting markers in lung cancer studies.

3 Methods

3.1 Training data

For training and evaluation, we gathered all manually curated protein sequences annotated with peptides or propeptides from UniProt Release 2022_01, excluding viral proteins and protein fragments from the query. All proteins that are annotated with a peptide that covers the full-length range of the mature protein were discarded, as these are not peptides in the sense of being proteolytically released from a precursor protein. The data was partitioned using GraphPart Teufel et al. [2023] into

5-fold at maximum 30\% Needleman–Wunsch pairwise identity between sequences in different folds. In total we analyzed 7623 protein sequences from 1503 biological species. To avoid a situation where different kinds of peptides are unevenly distributed between partitions, which could easily arise in the given dataset where the labeled classes are highly heterogeneous and unlikely to be generated by the same biological process, the clustering was performed prior to partitioning. The clustering was based on the flanking motifs of the peptides, as these are presumably the main causal feature that gives rise to peptide cleavage. To capture biochemical characteristics, we encoded flanking motifs using ESM-2.

3.2 Model architecture

We utilized DeepPeptide architecture Teufel et al. [2023] and tried to substitute the ESM-2 to other models: ESM-3, Ankh, ProtT5 (Fig. 1). We computed embeddings of the ESM-3 model in two ways: using protein sequences only and considering spatial structure of the corresponding proteins from AlphaFold.

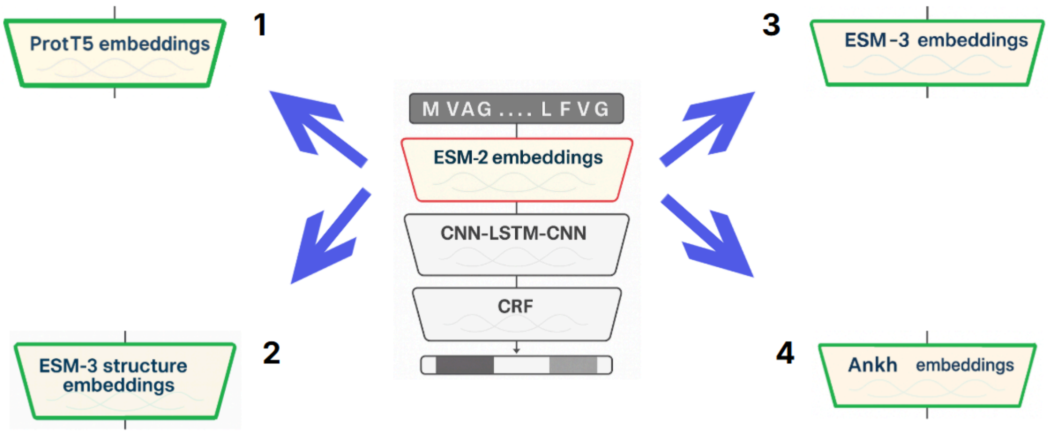

Fig. 1. Scheme of the model architecture modification.

The model is available at
https://github.com/spooky-soup/DeepPeptide/tree/main/prepare_data.

4 Results'

We trained the base DeepPeptide model represented by CNN, LSTM, another CNN and CRF blocks on data embeddings obtained with different models.
After evaluating metrics on the test dataset we discovered the original architecture to be more accurate in 7 cases out of 9. Experimentally obtained metrics on ESM-2 embeddings slightly differ from the ones provided in the article probably due to the differences in training process. All models were trained for 100 epochs.

Method	Peptides			Propeptides			All		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
ESM-2 (article)	0,69	0,43	0,53	0,64	0,46	0,54	-	-	-
ESM-2 (experiment)	0,44	**0,44**	**0,44**	**0,64**	0,43	**0,52**	0,53	**0,43**	**0,48**
ESM-3	0,39	0,41	0,40	0,62	0,45	**0,52**	0,50	**0,43**	0,46
Ankh	**0,50**	0,28	0,36	0,60	0,38	0,46	**0,55**	0,33	0,42
Prot T5	0,46	0,32	0,38	0,59	0,38	0,46	0,53	0,35	0,42
ESM-3 structure	0,33	0,36	0,34	0,59	**0,46**	**0,52**	0,45	0,41	0,43

Table 1. Model evaluation considering all proteins in the test

As human peptides are of a great biological interest, we additionally performed evaluation using human proteins only in the test. We intended to compare results of the model performance under two scenarios: (i) considering different species in the training set and (ii) restricting it to human (Table 2). However, training on human proteins only has failed due to the lack of the annotated data.

Embedding method	Peptides			Propeptides			All		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
ESM-2 (experiment)	0,24	0,22	0,22	0,66	0,45	0,53	0,49	0,37	0,42
ESM-3	0,28	**0,26**	**0,27**	**0,73**	**0,49**	**0,59**	**0,54**	**0,41**	**0,47**
Ankh	0,20	0,16	0,18	0,54	0,35	0,42	0,41	0,28	0,33
Prot T5	**0,29**	0,19	0,23	0,51	0,32	0,39	0,43	0,27	0,33
ESM-3 structure	0,20	0,16	0,18	0,68	**0,49**	0,57	0,49	0,36	0,42

Table 2. Model evaluation considering only human proteins in the test.

Additionally, we have checked whether the model used for encoding of the flanking regions affects clustering. We calculated embeddings using ESM-3 model and compared the results with clustering based on ESM-2 embeddings (Fig. 2).

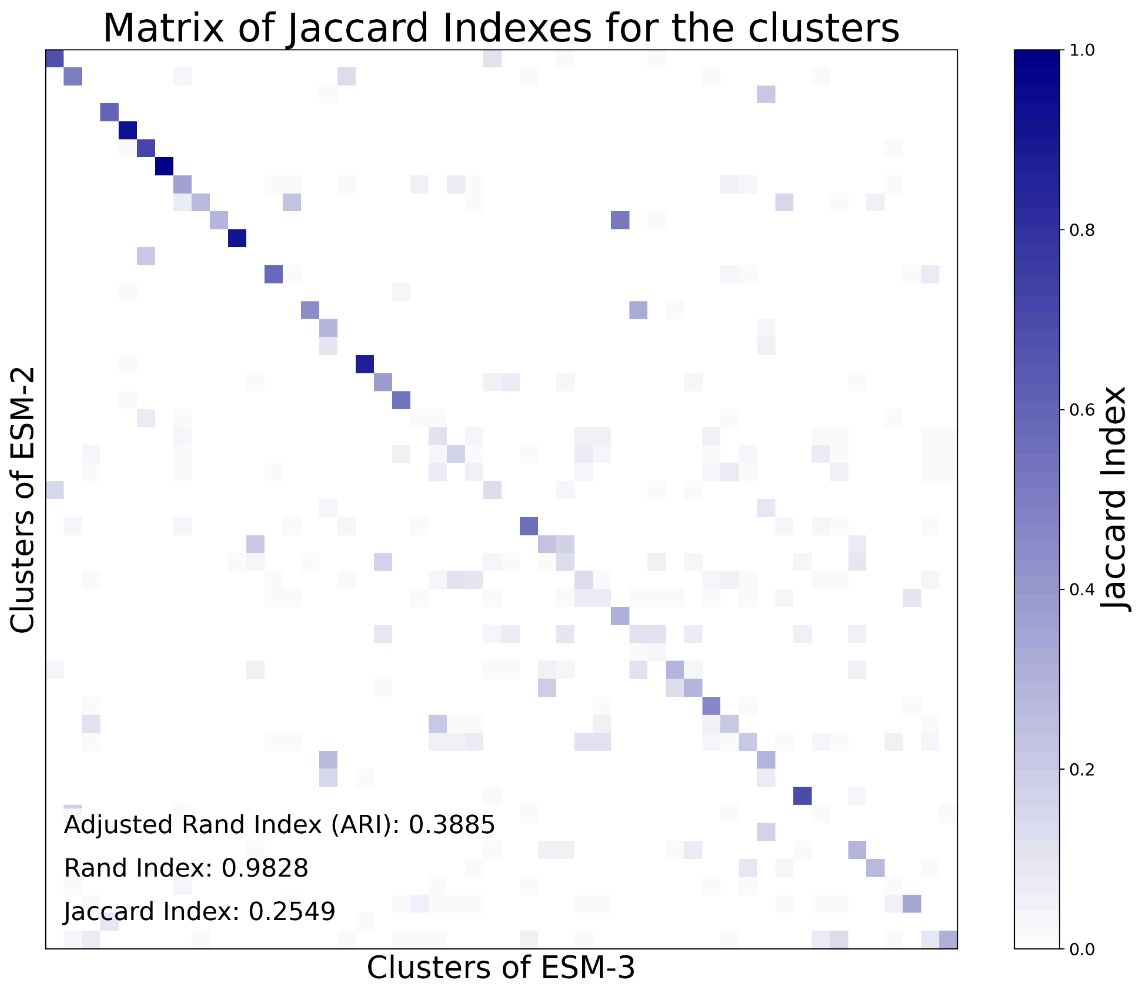

Fig. 2. Comparison of cluster structures

5 Conclusion and Future Work

ESM-3 and ProtT5 - based architectures outperformed baseline ESM-2 - based architecture while evaluating on human data only. Usage of spatial structure information for ESM-3 embeddings demonstrated poorer performance than embeddings did not show significant improvement in comparison to the We infer that the model switch does not severely affect the clustering of flanking motifs, however, other models can be considered in the future to make them more reliable. Alternative work directions may also imply dataset ablation to account for specificity of biological species and implementation of transformers in the model architecture.

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